HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed Python scripts for automating data extraction and transformation from various sources, reducing manual processing time by approximately **40%**.
- Engineered and maintained **Java Spring Boot** microservices, implementing REST APIs for core application functionalities and ensuring robust inter-service communication.
- Designed and implemented data models and ETL processes using **SQL** across PostgreSQL and Oracle databases for analytical reporting and business intelligence.
- Contributed to the enhancement of an enterprise-scale **document analysis** platform, focusing on improving information retrieval and text preprocessing pipelines.
- Optimized machine learning models for performance and scalability in production environments, incorporating **vector embeddings** and text analytics techniques.
- Translated research whitepapers into production-ready code for **AI architectures**, driving technical direction for document processing enhancements.
- Containerized applications using **Docker** and managed deployments on **AWS EC2** instances, ensuring efficient resource utilization and application availability.

SKILLS

AI/ML – Natural Language Processing (NLP), Information Retrieval, AI Architectures, Document Analysis, Model Optimization, Text Preprocessing, Vector Embeddings, Vectorization, Text Analytics, Optical Character Recognition (OCR), Machine Learning Models, Algorithmic Thinking

Programming Languages – Python, Java, SQL, R

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, Flask, FastAPI

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), Microsoft Azure, Google Cloud Platform (GCP)

DevOps & Tools – Git, Docker, Jenkins, Jira, Confluence, REST APIs, Microservices

Databases – PostgreSQL, MySQL, Oracle

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, Data Warehousing, Statistical Analysis

PROJECTS

Document Analysis Pipeline Enhancement

- Developed Python scripts to ingest and preprocess large volumes of unstructured documents for an enterprise document analysis system, applying **text preprocessing** and **vectorization techniques**.
- Integrated a newly researched algorithm for **information retrieval** into the existing platform, improving search result relevance by an estimated **15%** on a test corpus.
- Containerized the processing pipeline using **Docker** for consistent deployment across development and production environments on **AWS**.

AI Model Performance Optimization

- Fine-tuned a **NLP model** on a custom dataset of ~50,000 documents to improve performance for a specific domain, achieving a **0.78 F1-score**.
- Implemented optimizations for model inference speed using **FastAPI**, reducing average response time from 500ms to **210ms** for real-time document analysis.
- Collaborated with data science teams to define and implement metrics for evaluating model performance and identifying areas for further tuning.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)